

Lab 9 — Workflow Report

The Relics of the Void: 3D Content Generation Pipeline

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1. Image Generation (Lab 8)

The five booklet images were generated using Google Gemini (Nano Banana Pro model) with the Cinematic style preset. This model was selected over the lab-suggested SDXL-Lightning due to its superior photorealism and prompt adherence in 2026. Each image was deliberately designed as a single centred object on a plain dark background to optimise reconstruction quality in the subsequent TRELIS.2 stage. Prompts specified studio lighting, dark space background, and sharp focus to ensure clean foreground separation.

Generation Prompts

Relic I — Cryo-Core Crystal

"A single glowing blue-white alien crystal, hexagonal and translucent, with light pulsing from within, floating slightly above a surface, studio lighting, plain dark space background, sharp focus, photorealistic, 8k"

Relic II — Aethereal Helmet

"A single futuristic golden astronaut helmet, cracked reflective visor showing a colourful nebula reflection, floating in void, plain dark background, studio lighting, photorealistic, 8k, centered object"

Relic III — Alien Compass

"A single ornate alien navigation compass, circular, made of dark metal with glowing orange runes, lying flat, plain black background, studio product lighting, photorealistic, 8k, centered"

Relic IV — Stellar Seed

"A single bioluminescent alien seed pod, oval shaped, glowing green and purple, intricate organic surface patterns, floating, plain dark background, studio lighting, photorealistic, 8k, centered object"

Relic V — Obsidian Monolith

"A single black obsidian monolith tablet with glowing star map etched on surface, blue light emanating from engravings, standing upright, plain dark background, studio lighting, photorealistic, 8k, centered"

2. 3D Model Generation — TRELIS.2

Each booklet image was processed through Microsoft TRELIS.2, a 4-billion parameter feed-forward image-to-3D model using a novel O-Voxel sparse voxel representation with full PBR material output. TRELIS.2 was selected over Meshy and Tripo (limited free tiers), Tripo H3.1 (better suited to characters than hard-surface objects), and Rodin (paid). Settings used: 1024 resolution, 4096 texture size. GLB files were exported from the HuggingFace Space for direct import into Blender 5.0.1.

Model URL: <https://github.com/microsoft/TRELIS.2>

3. Blender Scene Construction

All five GLB models were imported into Blender 5.0.1 and arranged in a pentagonal formation at 2.8m radius on a dark reflective ground plane. A space galaxy image was applied as the world environment to provide ambient starfield lighting and ground reflections. Per-relic studio lighting was configured using accent spot lights with colour temperatures matching each relic's original booklet image. Emission shaders were applied to the crystal, monolith, and seedpod materials to

simulate internal glow effects. A camera orbit animation was keyframed across 240 frames at 7.5m radius and 3.8m height, rendered at 1920x1080 via Cycles (128 samples, denoising enabled).

Image model: Google Gemini — gemini.google.com | 3D model: Microsoft TRELIS.2 — github.com/microsoft/TRELIS.2